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Course: Calculus Beta

Assignment: MyLab Math - Uge 6

Find the Taylor series generated by f at $x = a$.

$$f(x) = e^x, a = 3$$

Let f be a function with derivatives of all orders throughout some interval containing a as an interior point. Then the Taylor series by f at $x = a$ is shown below.

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!} (x-a)^2 + \dots + \frac{f^{(n)}(a)}{n!} (x-a)^n + \dots$$

To find the Taylor series, begin by finding $f(3)$. Substitute 3 for x into the given function.

$$f(x) = e^x$$

$$f(3) = e^3$$

Now find $f'(3)$. First differentiate the given function.

$$f(x) = e^x$$

$$f'(x) = e^x$$

Find $f'(3)$. Substitute 3 for x into the first derivative of the given function.

$$f'(x) = e^x$$

$$f'(3) = e^3$$

Next find $f''(3)$. Differentiate $f'(x) = e^x$ to find the second derivative of the given function.

$$f'(x) = e^x$$

$$f''(x) = e^x$$

Now find $f''(3)$. Substitute 3 for x into the second derivative of the given function.

$$f''(x) = e^x$$

$$f''(3) = e^3$$

Find $f'''(3)$. First differentiate $f''(x) = e^x$ to find the third derivative of the given function.

$$f''(x) = e^x$$

$$f'''(x) = e^x$$

Now find $f'''(3)$. Substitute 3 for x into the third derivative of the given function.

$$f'''(x) = e^x$$

$$f'''(3) = e^3$$

Continuing in the same manner, you can see that each derivative of $f(x) = e^x$ will be the same, e^x , and each derivative evaluated at $a = 3$ will be e^3 . Thus, the derivative of the n th-term is e^x and derivative of the n th-term evaluated at 3 is e^3 .

Using this fact, type the missing terms of the Taylor series.

$$e^3 + e^3(x-3) + \frac{e^3}{2!}(x-3)^2 + \dots + \frac{e^3}{n!}(x-3)^n + \dots$$

Thus, the Taylor series generated by f at $x = 3$ is $\sum_{k=0}^{\infty} \frac{e^3}{k!} (x-3)^k$.